When Models Answer Without Seeing: Human-Calibrated Diagnosis of Linguistic Prior Exploitation in Vision-Language Models

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Abstract

Analogous to hypothesis-only baselines that exposed annotation artifacts in Natural Language Inference (Poliak et al. 2018), we investigate how much of VQA behavior is explainable without any visual information. Across multiple open-source Vision-Language Models (VLMs), blind inference reveals characteristic answer distribution biases shaped by the training distribution (McCoy, Frank, and Linzen 2024): models collapse to "no" for yes/no questions and "0" for count questions at rates far exceeding human baselines. To characterize how these priors depend on lexical cues, we introduce *control question variants*: four progressively weaker reformulations that remove entity anchors while preserving question type. A single instruction further reveals a spectrum from *instruction-gated* prior use to unconditional hallucination. Blind outputs are also highly overconfident: their token-confidence distributions remain high even for wrong committed answers, a calibration failure invisible to accuracy-only evaluations. Using a human blind-answer dataset (N=36, 113 questions, IRB-approved), we show that the model group closest to human blind priors is not the full VLM but the VLM backbone decoder: removing the vision encoder yields the highest human-model semantic agreement, while full VLMs and standalone LLMs exhibit distinct failure regimes. Question-type analysis then reveals where this overlap comes from: yes/no and action questions show strong human-model prior sharing, while count and grounded-reference questions are strongly anti-aligned.

1 Introduction

Vision-Language Models (VLMs) achieve impressive performance on visual question answering (VQA) benchmarks (Antol et al. 2015; Goyal et al. 2016), yet a basic question remains unanswered: how much of this performance reflects genuine visual grounding, and how much can be explained by linguistic priors inherited from large-scale language model pretraining?

This question parallels a well-known episode in Natural Language Inference (NLI): Poliak et al. (2018) showed that a model trained on hypotheses alone—ignoring the corresponding premises—substantially exceeded the random baseline across ten NLI datasets. The finding revealed annotation artifacts embedded in the benchmarks themselves. We apply the same diagnostic logic to VQA: the *image* plays the role of the premise, and the question is the hypothesis. A model that answers correctly from the question alone is exploiting a hypothesis-only shortcut.

McCoy, Frank, and Linzen (2024) provide the mechanistic explanation: large language models are shaped by their training objective over internet-scale text, and therefore produce outputs proportional to the probability of sequences in the training corpus. When a VQA question has a high-frequency answer in training data ("What color is the sky?" → "blue"), a VLM will produce that answer reliably regardless of the image. Blind VQA accuracy thus directly measures corpus frequency exploitation.

In this work, we conduct a diagnostic study of this phenomenon across multiple open-source VLMs. The paper is organized around three questions. First, how much blind accuracy can models obtain at scale, and what answer distribution biases underlie that performance? Second, how does prior-driven behavior change when we remove lexical anchors or explicitly instruct the model to answer from a plausible imagined scene? Third, how similar are blind model behaviors to human blind reasoning at the levels of correctness, answer distribution, and semantic agreement?

To answer these questions, we combine blind evaluation on 1,000 VQA questions, a control-variant perturbation suite, token-confidence analysis from output logprobs, and a human blind-answer dataset (N=36 participants, 113 questions). The goal is not to claim that blind answering is inherently illegitimate; the goal is to identify *which* priors are shared with humans and *which* are model-specific artifacts—a distinction that prior accuracy-centric diagnostics cannot make.

Main findings.

1. Models answer a substantial fraction of VQA questions correctly without any image input, confirming that blind prior exploitation is pervasive rather than anecdotal.
2. VLM backbone decoders (language backbone without the vision encoder) are the model group most closely aligned with human blind priors. They reach the highest human-model semantic agreement, while full VLMs are lower and standalone LLMs diverge further.
3. Full VLMs and standalone LLMs do not simply share a generic "model prior." They exhibit different failure regimes: full VLMs show stronger no-bias and zero-bias,

while standalone LLMs are more heterogeneous and often less human-aligned in answer semantics.

4. Models exhibit characteristic null-hypothesis biases under blind conditions: 62% of VLM yes/no answers are "no" (vs. human 33%), 70% of count answers are "0" (vs. human 5%), and color answers cluster on "black". These biases are model-specific—not shared with human blind reasoners.
5. Human-model prior alignment then explains where this split comes from: yes/no and action questions show strong overlap, while count and grounded-reference questions are anti-aligned. This identifies which priors are genuinely human-like and which are model-specific artifacts.
6. Blind outputs are severely overconfident: wrong committed answers remain in a high-confidence regime, and the instruction further raises confidence while reducing abstention, indicating more committed hallucination rather than more grounded reasoning.

Methodological contributions. Beyond the empirical findings, this work introduces two reusable diagnostic tools.

(i) A semantic control-question extraction pipeline. For each VQA question, we use GPT-4.1-mini with a structured JSON schema to automatically generate four progressively weaker reformulations: deictic markers removed, object noun replaced by a neutral placeholder, object replaced by its one-level-up hypernym, and content nouns fully pronominalized. This forms a controlled degradation axis that isolates *which lexical elements* carry prior information, independently of accuracy. The pipeline is fully specified (prompt and schema in Appendix C) and applicable to any VQA benchmark.

(ii) An operation–entity taxonomy for VQA prior analysis. We annotate each question with a semantic *operation type* (yes/no-existence, count, attribute, action, world knowledge, spatial, identity, temporal, text/OCR, comparison, causal) and an *entity type* (person, animal, object, food, vehicle, place, etc.). This taxonomy enables fine-grained diagnosis: yes/no and action questions show near-perfect human–model prior alignment (Jaccard $\approx$ 0.93), while count and OCR questions are anti-aligned ($r = -0.39$)—a distinction invisible at the aggregate level. Together, the two tools provide a principled diagnostic vocabulary for studying linguistic prior exploitation in any multimodal benchmark.

2 Related Work

Hypothesis-Only and Shortcut Baselines. Poliak et al. (2018) demonstrated that NLI classifiers achieve non-trivial accuracy using only the hypothesis, exposing annotation artifacts. Gururangan et al. (2018) further characterized these artifacts through systematic analysis of annotation bias. In VQA, Agrawal et al. (2018) showed that models relying on question-answer co-occurrence statistics could achieve high accuracy without visual understanding. Our work extends this diagnostic tradition by providing a human reference dataset alongside model evaluations, enabling calibrated measurement of the gap.

LMs as Corpus Frequency Models. McCoy, Frank, and Linzen (2024) showed that LLM failure patterns are predicted by three factors: the probability of the task, the input, and the output in the training corpus. Models are biased toward high-probability outputs regardless of task requirements. We operationalize this insight: blind VQA accuracy directly measures the probability of question-answer pairs in the pretraining distribution.

VLM Hallucination and Shortcut Learning. Hallucination in VLMs—confident generation of visually ungrounded content—has been widely documented (Ji et al. 2023; Rohrbach et al. 2018; Xiao and Wang 2021). Chen et al. (2024) proposed MMStar to audit visual grounding by filtering out questions solvable without images. VLind-bench (Lee et al. 2024) specifically measures language prior influence across diverse benchmarks. Han et al. (2024) identified visual shortcuts through spurious image features. Complementarily, **?** show that VLMs fail at fine-grained visual discrimination even when images are present—together with our work, bounding the visual grounding gap from both directions: models answer without images when they should not, and fail to use images when they must.

Human Cognition Under Uncertainty. Human reasoning under missing information is well characterized by dual-process theory (Tversky and Kahneman 1974; **?**): fast, heuristic System 1 thinking relies on representativeness and availability, while System 2 engages deliberate reasoning when cues are insufficient. Studying how humans answer questions without visual evidence thus belongs to a well-established research tradition in psycholinguistics and cognitive modeling—examining what mechanisms govern speculation under uncertainty (**?**). Critically, humans do not share the corpus-frequency bias of pretrained LLMs: their yes-bias (67% "yes") reflects pragmatic plausibility rather than training-distribution statistics, making human blind behavior the natural reference for what *linguistically natural* uncertainty looks like.

3 Experimental Setup

3.1 Models and Datasets

We evaluate open-source VLMs from the Qwen3-VL (Bai et al. 2025) and LLaVA (Liu et al. 2023, 2024) families, plus Qwen3 (Yang et al. 2025) LLMs as text-only baselines. The main blind-performance table reports the model subset for which the current zero-shot results are fully assembled; additional instruction-only comparisons are reported separately in Appendix E. All models are evaluated in zero-shot settings with temperature 0.

Our primary evaluation dataset consists of 1,000 questions sampled from the VQA v2.0 validation split (Goyal et al. 2016), spanning yes/no, numerical, and open-ended question types. For the human alignment analysis (Section 6), we use the 113-question subset on which the human study was conducted.

3.2 Blind Inference Conditions

We evaluate models under three conditions: (1) **Standard**: the original image with the question; (2) **Blind**: a blank 224×224 white PNG replacing the original image; and (3) **Blind+Inst**: blank image with an additional instruction appended to the question: "*Note: No images are provided. For each question, imagine an appropriate image exists and answer based on the most common or universal scenario.*"

3.3 Control Question Variants and Semantic Taxonomy

As a methodological contribution, we introduce two annotation layers applied to all 1,000 VQA questions.

Control question extraction. To characterize *how* models exploit linguistic priors, we generate four progressively weaker reformulations of each question using GPT-4.1-mini (temperature 0, seed 42, JSON schema mode) for reproducibility:

1. **deictic_removed**: Deictic markers (*this, that, here, there*) and spatial qualifiers (left/right/top/bottom) are removed while preserving fluency. Tests whether spatial anchoring drives shortcut behavior.
2. **object_removed**: The specific object noun is replaced with the neutral placeholder "object." Tests the contribution of entity identity to prior exploitation.
3. **weaker_object**: The specific noun is replaced with a one-level-up hypernym (dog → animal, car → vehicle, shirt → clothing). Tests whether category-level identity is sufficient to trigger prior recall.
4. **pronominalized**: All content noun phrases are replaced with minimal pronouns following grammar-aware rules ("What color is the dog?" → "What color is it?"; "How many cats are there?" → "How many are there?"). This maximally removes entity-specific prior cues while preserving question structure.

The degradation chain from original → `pronominalized` forms a monotonically decreasing axis of lexical prior cues, enabling a controlled measurement of how much each entity reference contributes to shortcut exploitation. The GPT extraction prompt and JSON schema are provided in Appendix C.

Operation–entity taxonomy. Each question is also annotated by the same GPT pipeline with two semantic dimensions: (1) *operation type*—the question's cognitive demand: yes/no-existence (`exist`), counting (`count`), attribute identification (`attr`), action recognition (`act`), world knowledge (`know`), spatial reasoning (`spat`), object identification (`ident`), temporal reasoning (`temp`), text/OCR (`text`), comparison (`comp`), and causal reasoning (`cause`); and (2) *entity type*—the primary visual referent: person, animal, object, food, vehicle, place, and other. These two dimensions cross-classify the 1,000 questions and enable the fine-grained alignment analysis in Section 6.3, where we show that human–model prior alignment varies dramatically across operation types (Jaccard 0.93 for yes/no vs. 0.13 for count), a distinction invisible at the aggregate level and unreachable without the taxonomy.

3.4 Human Data Collection

We recruited N=36 participants (15 men, 21 women; age 18–34) through online university community forums. All procedures were conducted under an IRB-approved protocol (compensation: 10,000 KRW/hour, consistent with standard rates). Participants answered 113 VQA-style questions in a text-only blind setting, relying only on the question text and general world knowledge. Responses could be given in Korean or English and were translated and normalized prior to analysis. After each answer, participants rated their confidence on a 5-point Likert scale. Unless otherwise noted, the human-comparison analyses in Section 6 are matched to the model condition used for those exports rather than assumed to be interchangeable across all blind prompts. Full protocol details are provided in Appendix A.

3.5 Metrics

We use four metric families, with different roles in the paper.

Task accuracy. We report **VQA accuracy** using the standard 10-annotator soft evaluation (Antol et al. 2015). This serves as context for how much blind success models obtain, but it is not our primary evidence for human-like prior structure. At the question level, human accuracy $h(q)$ is the proportion of 36 participants judged correct on question q; model accuracy is binary at the single-answer level.

Answer-distribution overlap. To measure whether humans and models converge on the same blind answers even when they are not always correct, we compute **Jaccard overlap of the top-3 answers** per question. For human answer set H_q and model answer set M_q, this is:

$$\mathrm{Jaccard}(q) = \frac{|H_q \cap M_q|}{|H_q \cup M_q|}.$$

This metric captures shared prior structure independently of exact correctness.

Agreement and semantic similarity. For the answer-agreement analyses in Section 6, we compare human-human, human-model, and model-model answer pairs after shared answer normalization. We use two classes of agreement metrics.

First, **lexical metrics**: **exact match**, **token Jaccard**, **ROUGE-1**, and **chrF**. These quantify increasingly soft surface overlap, from exact normalized identity to word- and character-level overlap.

Second, **semantic metrics**: **Sentence-BERT cosine** (Wang et al. 2020), **SimCSE cosine**, and **BERTScore F1**. These are the main metrics for our central claim, because the question is not whether models reproduce the exact human string, but whether they recover the same human prior at the level of meaning. In the main text, Sentence-BERT is the primary semantic metric; the others are used as robustness checks.

Alignment and reliability. To compare human and model difficulty patterns across questions, we compute **Pearson difficulty correlation**, namely Pearson r between per-question human accuracy $h(q)$ and per-model accuracy. We prefer Pearson over rank-based alternatives because the comparison is over proportions and the relationship is approximately linear in the current analyses. To quantify group consistency, we also report **Krippendorff's** α (nominal scale) for within-group and cross-group inter-rater reliability.

Confidence. Finally, we measure a **token-confidence proxy** from generation logprobs (top-5 token logprobs, temperature 0). For each answer, we convert each generated token logprob to probability and average across generated tokens. This produces a mean token probability in $[0, 1]$, which we use to characterize whether blind answers are cautious, abstaining, or confidently hallucinated.

4 Linguistic Prior Exploitation: Distribution Biases and Prior Structure

We characterize the *structure* of blind model behavior—what answer distributions models produce, and how those distributions depend on lexical cues in the question. Accuracy serves as secondary context; the primary evidence is the shape of answer distributions and how they compare to human baselines.

Blind VQA accuracy ranges from 35–49% across models (see Appendix D for the full per-model breakdown). Text-only Qwen3 LLMs (no vision encoder) achieve 35–38%, approaching their VLM counterparts, confirming that blind performance is driven by the language backbone. Scale does not monotonically improve blind VQA accuracy, consistent with a ceiling on high-frequency question–answer patterns. The instruction has little effect on accuracy even when it substantially changes answer content (see Appendix E), confirming that prior-driven answering is not corrected by awareness of the missing image.

4.1 Answer Distribution Biases

A key question is not just *whether* models answer correctly under blind conditions, but *how* their answer distributions differ from humans and from ground truth. As shown in Figure 1, three systematic biases emerge:

No-bias in yes/no questions. Under blind conditions, 70% of model yes/no answers are "no" (vs. ground truth: 47% no). Humans show a smaller bias (53% no), suggesting that the model's extreme no-bias reflects an epistemic default—the null answer under uncertainty—rather than a corpus prior. Notably, this reverses under blind+inst ("yes" increases from 30% to 55%), confirming that the "no" default is instruction-suppressed rather than a genuine linguistic prior.

Zero-bias in count questions. 75% of model numeric answers under blind conditions are "0," while humans produce a much more distributed count distribution. The interpretation is consistent: in the absence of visual evidence, models default to "there are none."

Black-bias in color questions. Among open-ended color answers, "black" accounts for 16% of model responses under blind conditions, far exceeding its ground-truth frequency. This likely reflects the modal color in the training distribution for generic object descriptions.

These null-hypothesis biases illustrate that models are not randomly sampling from their prior—they produce a specific failure mode that partially mimics the structure of a reasonable uncertainty response.

5 Characterizing Prior Reliance

The analyses in this section move from *how much* blind performance models obtain to *how* they produce it: whether their priors survive lexical degradation, whether a small instruction changes their willingness to commit, and whether those commitments are calibrated. Figure **??** shows the human-alignment pattern by model group and scale, and Figure 3 shows how that alignment weakens under control-variant degradation.

5.1 Scale and Control Question Degradation

Figure **??** previews the main architectural result from a scale perspective: backbone decoders remain closer to humans than full VLMs or standalone LLMs across parameter ranges, and simple scaling does not erase that gap. The control-variant perturbations then show that even the stronger human-aligned groups rely on lexical anchors. As entity-specific content is progressively removed—from the original question (C) to a hypernym-weakened variant (B) to a fully pronominalized form (A)—human-model semantic alignment drops for every group, confirming that blind success is carried by entity-specific prior signals rather than abstract reasoning alone.

5.2 Instruction-Gated Hallucination

A key question is whether models' blind answers reflect a readily available prior that is always active, or a latent prior that requires explicit permission to express. We investigate this by comparing model outputs between the **Blind** and **Blind+Inst** conditions.

We classify model outputs in the blind condition as: *hard-abstained* (explicit refusal: "I cannot answer", "no image"); *soft-abstained* (hedging outputs: "nothing", "none", "unanswerable", "nowhere"); *hallucinated_correct* (specific answer matching GT); or *hallucinated_wrong* (specific answer not matching GT). Hard abstentions are near-zero across all models (0.0–0.3%), consistent with the baked-in "Answer using a single word" instruction suppressing explicit refusals. The real signal is *soft abstention*: 11–19% of blind outputs are hedging responses, varying by model family.

Figure 4 shows the *soft abstention collapse rate*: the fraction of soft-abstained blind responses that convert to specific answers under the Blind+Inst instruction. This rate varies dramatically:

Model	**Soft-abs (blind)**	**Collapse rate**
Qwen3-VL-8B	12.9%	71.9%
LLaVA v1.6-Mistral	15.8%	51.9%
LLaVA v1.6-Vicuna	11.3%	28.3%
LLaVA v1.5	18.7%	17.1%

We interpret this spectrum as measuring the degree to which prior activation is *instruction-gated*. Qwen3-VL, a highly instruction-tuned model, largely suppresses its prior under blind conditions but rapidly activates it when given

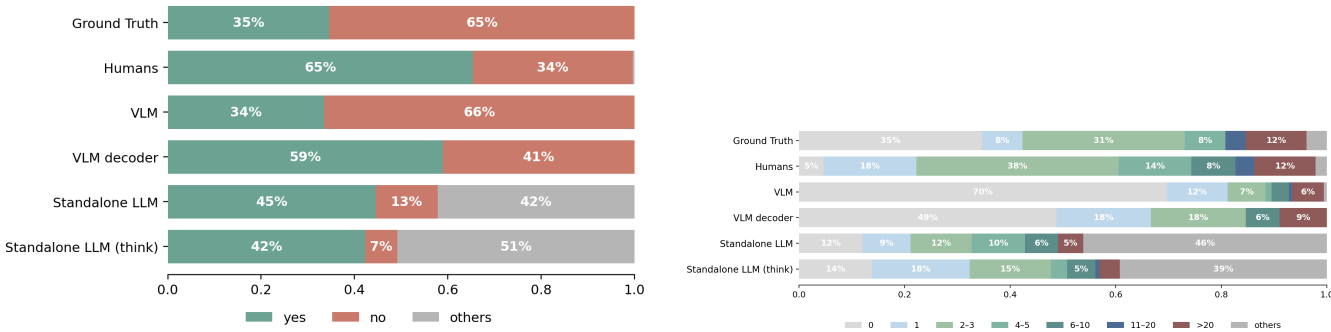

Figure 1: Answer distribution biases under the instructed blind condition on the human-study question subset ($q = 26$ yes/no, $q = 26$ number; $h = 40$ participants). **Left (yes/no):** VLMs still overproduce a default answer pattern relative to humans. **Right (count/number):** model groups remain collapsed toward small default counts, especially "0," while humans distribute across small integers. These divergences confirm that blind answers are structured but not simply human-like.

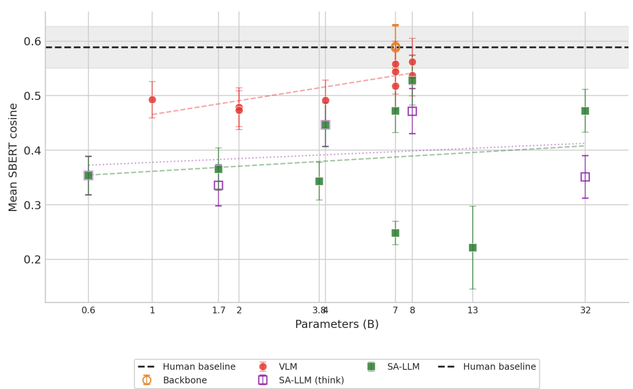

Figure 2: Human–model SBERT agreement versus model scale by model group on the inst_blind variant-C human-comparison subset. Alignment does not improve monotonically with parameter count; backbone decoders sit closest to the human reference across scales, indicating that architecture matters more than raw size.

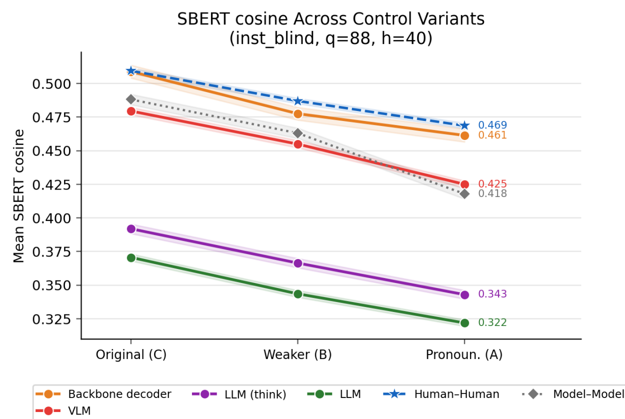

Figure 3: Human–model SBERT agreement across control variants (Original C $\rightarrow$ Weaker B $\rightarrow$ Pronominalized A) on the free-text human-comparison subset ($q = 88$, $h = 39$). All model groups lose alignment as entity anchors are removed. The human–human ceiling (dashed) drops only slightly, confirming that the degradation is model-specific. Full results for all metrics (chrF, ROUGE-1, BERTScore, SimCSE, Jaccard, exact match) are in the supplementary heatmaps.

permission. LLaVA v1.5, by contrast, hallucinates *unconditionally*: it produces specific answers whether or not instructed to, and shows almost no response to the permission instruction. This distinction is mechanistically important: instruction-gated models have a functioning "uncertainty" signal that instruction can override, while unconditional models lack this gate entirely.

Additionally, 33.6% of all answers (not just soft-abstentions) change between blind and blind+inst conditions for Qwen3-VL-8B—a large behavioral shift driven by a single sentence. The yes/no distribution shifts from 70% "no" in blind to 55% "yes" in blind+inst, consistent with the instruction activating a "plausible scenario exists" prior that overrides the default epistemic null.

5.3 Confidence Calibration

Confidence analysis reveals that blind hallucinations are produced with high token-level confidence rather than cautious uncertainty. The instruction does not merely change *which* answer is produced; it shifts mass away from abstentions and toward committed outputs that remain highly confident at the token level.

Soft-abstained outputs occupy a lower-confidence regime than committed answers, confirming that hedging responses reflect genuine uncertainty. However, the gap is modest, and the baseline level of confidence even for wrong committed answers remains high. The primary failure is therefore not underconfidence but confident hallucination.

Quantifying this overconfidence with Expected Calibration Error and testing the corpus-frequency correlation (McCoy, Frank, and Linzen 2024) directly are reserved for future work.

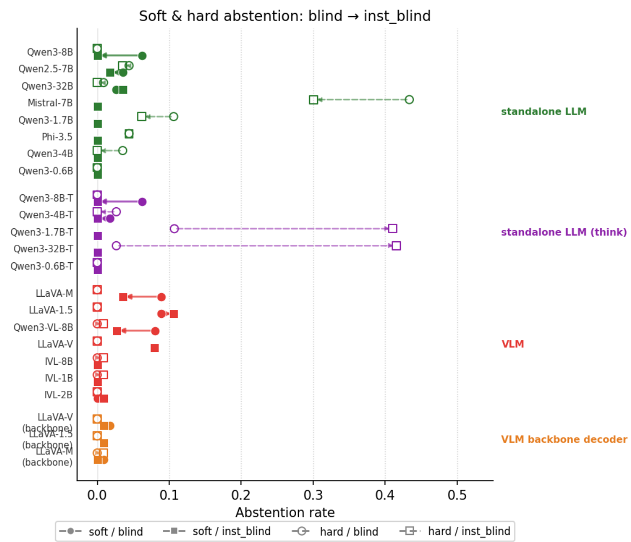

Figure 4: Response behavior under Blind and Blind+Inst for variant C on the human-study subset. The instruction primarily reduces abstention and converts hedged outputs into committed answers rather than improving grounding. This shift is strongest for VLMs and standalone LLMs, while backbone decoders are comparatively stable.

6 Human-Model Alignment

The central result of this section is architectural: when models answer VQA questions without images, the group closest to human blind responses is the VLM backbone decoder, not the full VLM. VLM backbone decoders (the language backbone extracted without the vision encoder) reach the highest human-model SBERT similarity (0.501), within 0.067 of the human-human ceiling (0.568). Full VLMs are lower (0.480), while standalone LLMs span a much wider range.

The key point is therefore architectural rather than purely descriptive: the model component that best recovers human blind priors is the language backbone exposed in the decoder-only setting, not the full vision-language stack. The entity-type analysis that follows explains where this alignment is strongest.

6.1 Humans as a Reference for Prior Structure

Human participants answering without images still rely on world knowledge and linguistic priors, but they are not shaped by the same corpus-frequency objective that drives pretrained LLMs. This makes human blind behavior a reference for what *linguistically natural* uncertainty looks like—as opposed to what the training distribution would predict.

The contrast is sharpest in answer distributions (Figure 1). For yes/no questions, VLMs default to "no" 62% of the time, while humans lean toward "yes" 67%—the opposite direction. For count questions, 70% of VLM answers are "0" while humans produce a distributed count distribution (modal answer: 2, followed by 1 and 3). This reversal is not a calibration gap in the same direction; it indicates that the model prior is qualitatively different from the human prior

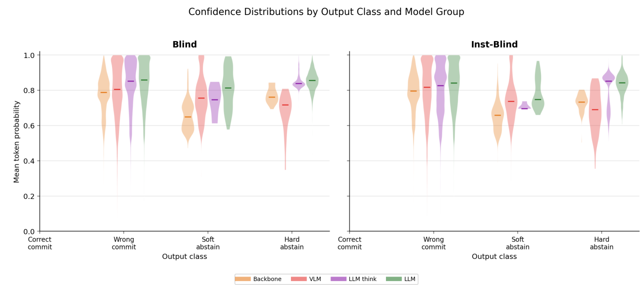

Figure 5: Distribution of mean token probability by output class under Blind and Blind+Inst on the human-study subset ($q = 113$, $h = 40$), shown as violin distributions by model group. Wrong committed answers remain in a high- confidence regime, while abstentions occupy a lower-confidence regime. The instruction shifts outputs from abstention toward confident commitment rather than toward calibrated uncertainty.

for these question types.

6.2 Which Models Are Most Human-Like?

Not all model families align equally with human blind behavior. We compare four groups—VLMs, VLM backbone decoders (the same language backbone without the vision encoder), standalone LLMs, and thinking-mode LLMs—using both difficulty correlation and pairwise answer similarity. The main interpretive question is whether the model component that best matches humans is the visual stack, the language backbone, or the instruction-tuning regime layered on top.

Pairwise answer similarity. We compute pairwise SBERT cosine similarity (Wang et al. 2020) between all answer pairs (human-human, human-model, model-model) across the 374 free-text control questions. Human-human agreement provides the ceiling: $\overline{\text{SBERT}}_{\text{HH}} = 0.568$. VLM backbone decoders reach 0.501—closest to the human ceiling—while full VLMs score 0.480, standalone LLMs 0.398, and thinking-mode LLMs 0.426. The gap between backbone decoders and full VLMs ($\Delta = 0.021$) is consistent across lexical metrics (exact match, chrF) and semantic metrics (SimCSE and BERTScore), indicating that removing the vision encoder does *not* hurt human alignment and may even improve it by reducing the model's reliance on spurious visual features.

Difficulty correlation by model group. Pearson correlation between per-question human accuracy $h(q)$ and per-model accuracy further separates the groups. Under blind conditions, VLMs correlate at $r = 0.50$–0.62; backbone decoders at $r = 0.58$–0.68; standalone LLMs at $r = 0.35$–0.55. The Blind+Inst condition consistently raises correlations by up to $\Delta r = +0.10$ for VLMs, narrowing the gap to the human reference. This pattern is consistent with the instruction-gated hallucination result (Section 5.2): receiving the "imagine the scene" instruction pushes models to activate the same distributional priors that drive human performance on these questions.

Effect of scale on human alignment. Within-family scale trends are comparatively weak: human–model SBERT alignment does not improve monotonically with parameter count, indicating that alignment with human blind priors is determined more by architecture and instruction-tuning regime than by raw scale. Detailed scale plots are provided in the Appendix.

Inter-rater structure. Supplementary inter-rater heatmaps show that human participants cluster separately from all model groups, confirming a qualitative difference in reasoning strategy rather than a mere accuracy gap. Backbone decoders sit closest to the human cluster, standalone LLMs furthest. Krippendorff's α decreases when any model is added to the human rater pool (from 0.554 to 0.535–0.543), but the decrease is smallest for backbone decoders, reinforcing that the language backbone is the primary driver of human-like prior structure in VLMs.

6.3 Shared vs. Divergent Priors by Entity Type

Not all questions trigger the same degree of human-model prior alignment. The entity taxonomy introduced in Section 3.3 provides the analytical structure: entity groups form coherent semantic categories in the question set, making entity type a principled grouping dimension for alignment analysis.

We measure **difficulty correlation**—Pearson r between per-question human accuracy $h(q)$ and VLM accuracy—within each of the 9 entity types across 113 questions. Figure 6 shows the distribution of questions by entity type (left) and the resulting correlations (right).

Three alignment regimes emerge:

Strong alignment (animal, place). Animal and place questions show near-perfect human–model difficulty correlation ($r = 0.90$, 0.91). Both humans and models exploit the same entity-specific corpus frequencies for these categories—questions about animals or locations tend to have predictable answers that align with statistical regularities in training data.

Moderate alignment (food, person, other). Food ($r = 0.77$) and person ($r = 0.64$) entities show moderate alignment. Models capture the broad easy/hard structure of human difficulty but diverge on ambiguous cases, particularly for person questions where identity and contextual knowledge interact.

Divergence (object, product). Object questions show weak alignment ($r = 0.13$), reflecting the high diversity of object categories that span many prior strengths. Product questions are anti-aligned ($r = -0.48$): when humans succeed on product queries, models tend to fail, likely because product-specific prior knowledge is sparser in model training data than in human experience.

The per-question scatter and quadrant-style diagnostics are provided in the Appendix. They show that shared priors are strongest when both parties are either similarly uncertain or similarly knowledgeable, while count and product questions drive many of the strongest divergences.

6.4 Human-Model Divergence in Answer Patterns

We measure human-model alignment using two complementary metrics. First, we compute **Pearson difficulty correlation**: for each question q, we compute the human group accuracy $h(q)$ (proportion of 36 participants correct) and correlate it with per-model accuracy. We prefer Pearson over Spearman here because the relationship is approximately linear (max $|r - \rho| = 0.026$ across all conditions), and operating on proportions avoids the information loss from rank conversion on binary data. Second, we compute **Krippendorff's** α (nominal scale, binary correctness) to measure within-group and cross-group reliability.

Human difficulty estimates are highly reliable: $\mathrm{ICC}(2, 36) = 0.963$ across 36 raters, justifying use of $h(q)$ as a stable difficulty signal. Within-group human reliability is $\alpha_{\text{human}} = 0.554$ (moderate, appropriate for a genuinely ambiguous task without visual input).

Pretrained VLMs show moderate correlation with human difficulty under blind conditions (Pearson $r = 0.50$–0.62 depending on model), confirming that human and model difficulty patterns partially overlap—both exploit linguistic priors to some extent. However, adding any model to the human rater pool *decreases* α (from 0.554 to 0.535–0.543), confirming that models are structurally less consistent with the human group than humans are with each other.

In the current analysis, Blind+Inst often yields slightly higher human-difficulty correlations than Blind alone for several model families. This pattern is consistent with the soft-abstention-collapse result in Section 5.2, but it should be treated as provisional until the human comparison conditions are fully matched across settings.

Appendix heatmaps quantify the same structural separation from a different angle: human-human similarity exceeds all cross-group scores, humans form a distinct cluster, and backbone decoders sit closest to that cluster. The pattern is consistent across lexical and semantic metrics.

Semantic similarity analysis reinforces this finding: model answers have higher embedding similarity to ground truth (mean 65.8 vs. human 64.2) despite being generated without visual evidence, and the human similarity distribution has a longer left tail (more diverse wrong answers), consistent with exploratory reasoning rather than prior retrieval.

7 Discussion

The instruction-gated spectrum as an architectural insight. The dramatic variation in soft-abstention collapse rates (17% for LLaVA v1.5 vs. 72% for Qwen3-VL) suggests that instruction-following capability plays a role in prior activation that goes beyond accuracy. Instruction-tuned models have learned to gate prior expression behind contextual permission signals, while models with weaker instruction following leak priors unconditionally. This is not simply a capability difference—LLaVA v1.5 is competitive on

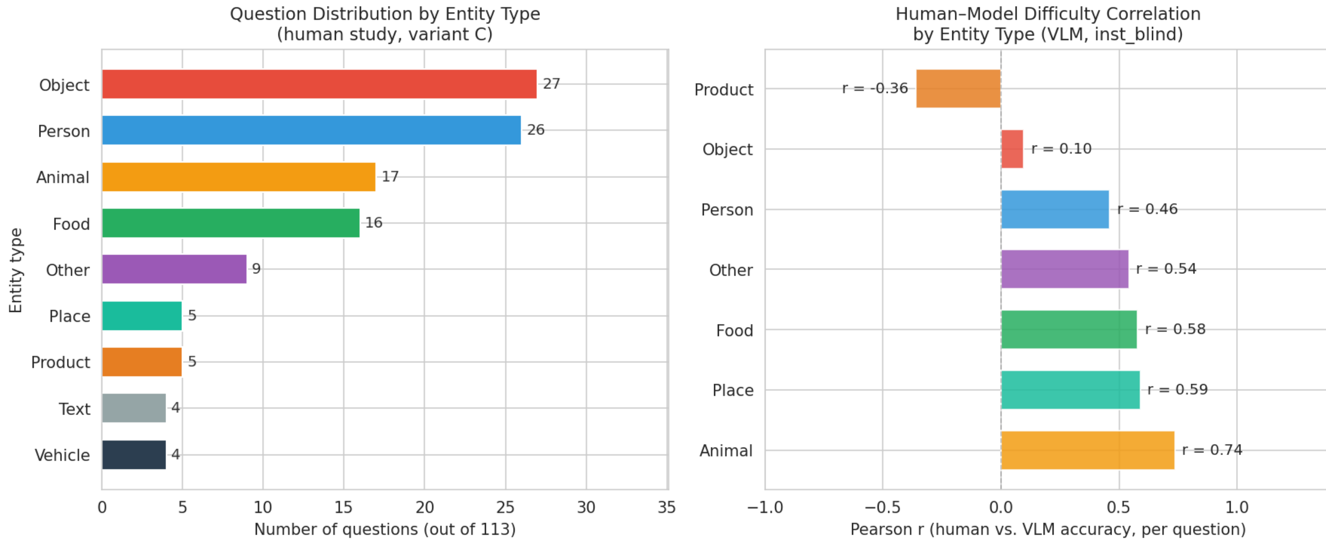

Figure 6: **Left:** distribution of 113 study questions by entity type. Object and person dominate; place, product, text, and vehicle have small samples ($n \leq 5$). **Right:** Pearson r between per-question human accuracy and VLM accuracy (inst_blind, variant C) within each entity type. Animal ($r = 0.90$) and place ($r = 0.91$) questions show near-perfect human–model alignment; food and person show moderate alignment ($r = 0.77, 0.64$). Object questions show weak alignment ($r = 0.13$), while product questions are anti-aligned ($r = -0.48$), indicating that human and model difficulty patterns diverge for product-related queries.

standard VQA accuracy—but a behavioral difference in how priors are accessed.

Implications for benchmark design. Our results suggest that any VQA benchmark can be partially solved by linguistic priors alone, and that the magnitude of this effect varies predictably with question type, answer distribution, and entity specificity. We propose that blind VQA evaluation should become a standard diagnostic for new VQA benchmarks: a high blind accuracy (>40%) is at least a warning sign that the benchmark may over-index on knowledge-accessible question types.

Limitations. Our human study involves 36 participants recruited from university communities in a single language (Korean/English), limiting cultural and demographic generalizability. Model evaluation uses greedy decoding (temperature 0), which may underestimate answer diversity and overestimate confidence. The control variant GPT extraction uses a best-effort seed (seed is advisory in the Responses API) and results may vary slightly across runs. Human data was collected under the Blind+Inst condition only, so direct comparisons with model blind (no instruction) behavior rely on the assumption that the instruction has a comparable normalizing effect on humans and models—an assumption our instruction-gated analysis partially supports but does not fully validate. A direct corpus-frequency correlation—testing whether per-question blind accuracy predicts answer frequency in the VQA v2 training distribution—would strengthen the mechanistic account (McCoy, Frank, and Linzen 2024) and is deferred to future work.

8 Conclusion

We present a human-calibrated diagnostic study of linguistic prior exploitation in Vision-Language Models. Analogous to hypothesis-only baselines in NLI, blind VQA reveals that VLMs produce characteristic answer distribution biases shaped by training-corpus frequencies—and that these biases are only partially shared with human blind reasoners. We introduce control question variants to probe how lexical cues drive prior activation, and use token-confidence analysis to expose the severe overconfidence that accompanies blind hallucination. The instruction-gated spectrum reveals a behavioral axis across model families that is invisible to accuracy-only evaluation.

The central finding of the human alignment study is that prior sharing is question-type-specific. Yes/no and action questions show near-complete human-model overlap (Jaccard ≈ 0.93), while count and grounded-reference questions are anti-aligned (difficulty $r = -0.39$). The model zero-bias and no-bias are not linguistic priors in any human sense—they are model-specific failure modes. At the architecture level, VLM backbone decoders (language backbone without vision encoder) achieve near-human pairwise answer similarity (0.501 vs. human 0.568), and their yes/no distribution reverts to human-like rates when the vision encoder is removed, confirming that the no-bias is a VLM artifact rather than a corpus-frequency prior. These findings argue that diagnosing prior exploitation requires attending to both the language backbone and the vision–language integration layer separately.

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A Human Study Protocol

A total of 36 participants (15 men and 21 women; age range: 18–34 years) were included in the study. Participants were recruited via online student community forums at universities. All procedures were conducted under an IRB-approved protocol and adhered to the principles of the Declaration of Helsinki. Informed consent was obtained from all participants prior to participation.

Each participant completed 113 VQA-style questions. Participants were compensated at a rate of 10,000 KRW per hour, consistent with standard compensation for similar annotation tasks.

For VQA, participants were instructed to provide free-form answers that were as concise as possible. Responses could be given in either Korean or English and were translated and processed using the same answer normalization and aggregation strategy described in Appendix B.

B Answer Aggregation

We cluster semantically equivalent answers (e.g., "3" and "three") into canonical forms using GPT-4o mini (OpenAI 2025). The prompt instructs the model to group synonymous responses, select the most concise canonical form, and return results in structured JSON. Human responses are then aggregated into a confidence-weighted distribution over canonical answers, where each answer's weight is the sum of confidence scores from all annotators selecting that answer, normalized to sum to 1:

$$H(a_k^*) = \frac{\sum_{i:\phi(a_i)=a_k^*} c_i}{\sum_i c_i} \tag{1}$$

where c_i is participant i's Likert confidence (normalized to $[0,1]$), ϕ maps raw answers to canonical answers, and K is the number of canonical answer clusters.

C Control Question Extraction Prompt

Control variants are extracted using GPT-4.1-mini-2025-04-14 (temperature 0, seed 42) with strict JSON schema mode for reproducibility. The system prompt instructs the model to generate all four variants in a single API call per batch of 50 questions. The schema enforces: {deictic_removed, object_removed, weaker_object, pronominalized} as required string fields, with additionalProperties: false. For pronominalized, the prompt specifies grammar-aware substitution rules to ensure natural pronominalization ("What color is it?" not "What color is the object?").

D Blind VQA Accuracy Summary

E Blind Awareness Instruction

After the original prompt and the blank image, models in the Blind+Inst condition receive: *"Note: No images are provided. For each question, imagine an appropriate image exists and answer based on the most common or universal scenario."*

Table 2 shows accuracy comparison between Blind and Blind+Inst conditions on VQA. The instruction usually changes accuracy by only a few points, confirming that it does not reliably improve performance even when it substantially changes answer behavior.

F Supplementary Per-Question and Scale Diagnostics

The main text emphasizes group-level results. This appendix collects the per-question and scale-sensitive views that support the same claims but are too dense for the main paper.

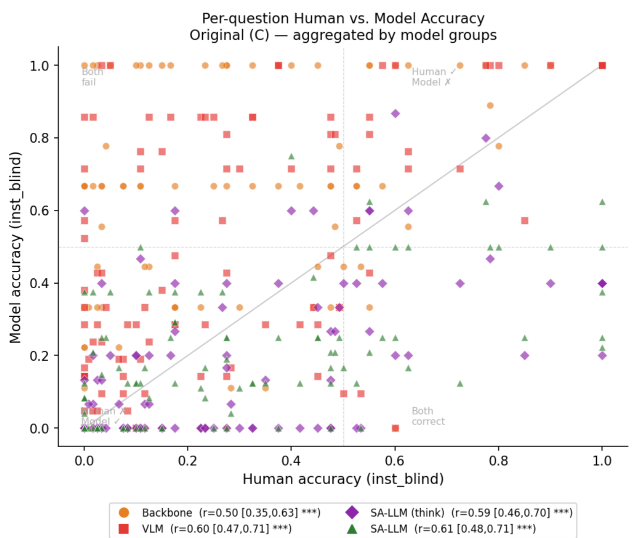

Figure 7: Per-question human accuracy versus model accuracy on the instructed blind human-study subset ($q = 113$, $h = 40$), faceted by model group. This raw scatter underlies the aggregated control-variant trends in Figure 2.

G Answer Distribution by Model Group (Notebook 13)

Figure 10 repeats the blind-condition answer-distribution comparison from the main text. A fuller notebook-driven breakdown by all model groups and both Blind / Blind+Inst conditions remains part of the analysis pipeline, but the current main paper claims only require the VLM-versus-human contrast shown here.

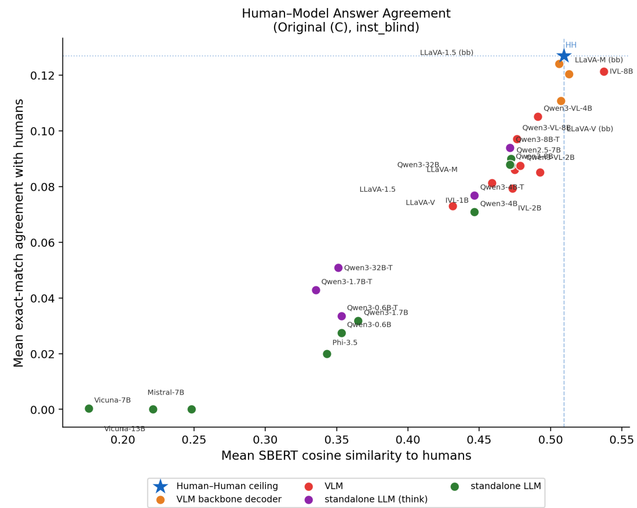

Figure 8: Per-model human–model agreement on the free-text subset ($q = 88$, $h = 39$). X-axis: mean SBERT similarity to human answers. Y-axis: mean exact-match agreement. This figure makes the same group separation as the main text visible at model granularity.

H Agreement by Control Variant — All Metrics

Figures 11–21 extend Figure 3 from §5.1 to all remaining agreement metrics. Each lineplot shows mean human–model agreement per model group across the three control variants (Original C $\to$ Weaker B $\to$ Pronominalized A), with the human–human ceiling (dashed blue) as reference. Each heatmap summarises the same data as a group$\times$variant grid. The consistent downward trend across all metrics confirms that entity removal reduces human–model alignment independent of the metric used.

I SBERT Agreement by Control Variant — Group Heatmaps

Figure 23 provides the heatmap counterpart to the SBERT lineplot shown in Figure 3 (§5.1). Rows correspond to model groups and the human–human baseline; columns to the three control variants.

J Inter-rater Agreement Heatmaps — All Metrics

Figures 24–37 summarize the inter-rater agreement structure for variant C under the inst_blind condition. In each figure, the top panel shows group-mean agreement and the bottom panel shows the full per-rater matrix (36 humans + all model groups). The block structure is consistent across metrics: humans cluster tightly together, backbone decoders agree more

Table 1: Zero-shot blind VQA performance of humans and models. Y/N = yes/no accuracy; Num = numerical; Other = open-ended (Accuracy/Similarity). Bold: best; underline: second best.

Model	**LLM**	**Size**	Y/N	Num	Other	Avg
Humans (N=36)	–	–	66.6	13.8	13.0/45.5	39.0
LLM Baseline (text-only)						
Qwen3	Qwen3	4B	54.9	19.0	24.7/55.2	38.8
		8B	50.3	17.1	22.1/53.2	35.3
VLM (blind)						
Qwen3-VL	Qwen3	2B	71.8	17.1	23.8/55.5	46.4
		4B	71.5	19.0	28.9/56.4	48.0
		8B	71.5	21.9	27.0/54.9	**48.0**
LLaVA v1.5	Vicuna	7B	70.3	17.1	19.4/48.4	43.8
LLaVA v1.6	Mistral	7B	69.6	20.0	27.0/55.0	47.0
LLaVA v1.6	Vicuna	7B	74.4	20.0	27.2/56.2	49.4

Table 2: VQA accuracy comparison with and without blind instruction. $\Delta_{\text{inst}} = \text{Blind}_{\text{inst}} - \text{Blind}$.

Model	GT	Blind	Δ
InternVL3.5-1B	80.5	43.6	0.2
InternVL3.5-2B	82.1	45.5	-0.2
InternVL3.5-4B	83.0	45.0	2.2
InternVL3.5-8B	86.9	47.1	-1.7
Qwen3-VL-2B	83.0	45.0	0.7
Qwen3-VL-4B	85.9	45.5	1.9
Qwen3-VL-8B	89.4	44.3	3.1

with humans than full VLMs do, and standalone LLMs (especially thinking-mode variants) occupy a distinct region.

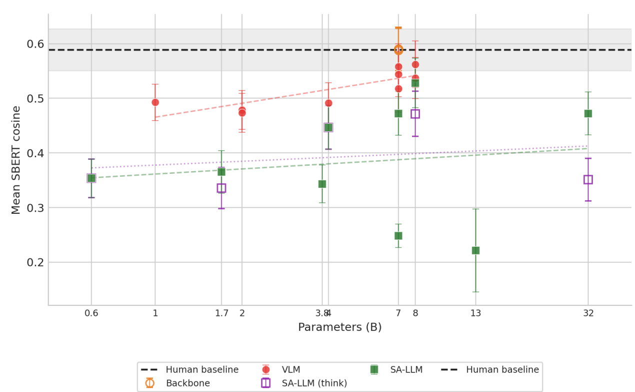

Figure 9: Human–model SBERT agreement versus model scale by model group on the inst_blind variant-C human-comparison subset. Scale effects are weaker than the architectural differences emphasized in the main text.

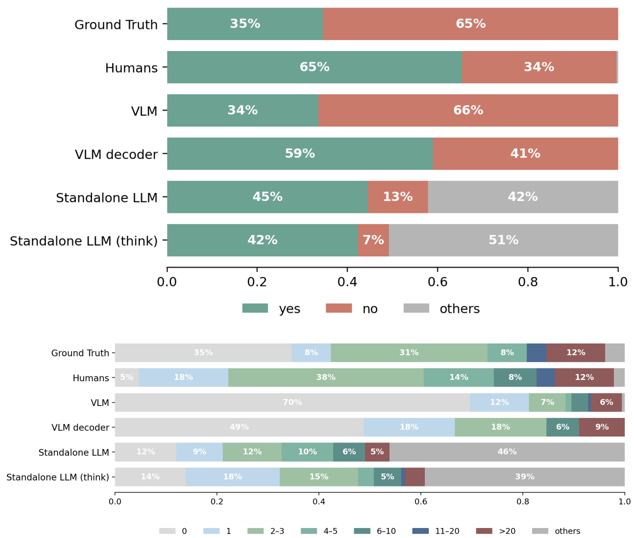

Figure 10: Answer-distribution biases under the instructed blind condition across all model groups. Top: yes/no answers. Bottom: count answers. The descriptive patterns remain structured, but matched-size and model-group comparisons are best interpreted as robustness checks rather than as the main architectural evidence.

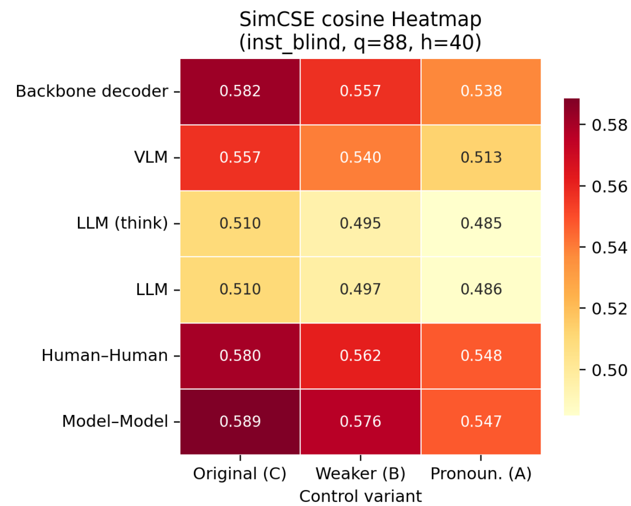

Figure 12: SimCSE agreement heatmap (group $\times$ variant, HM pairs).

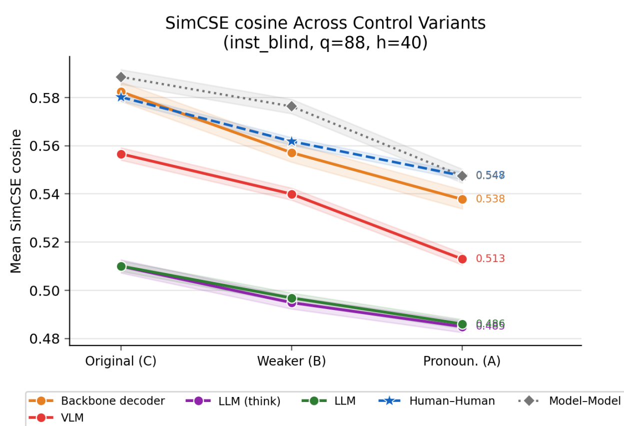

Figure 11: Human–model SimCSE cosine agreement across control variants (by model group).

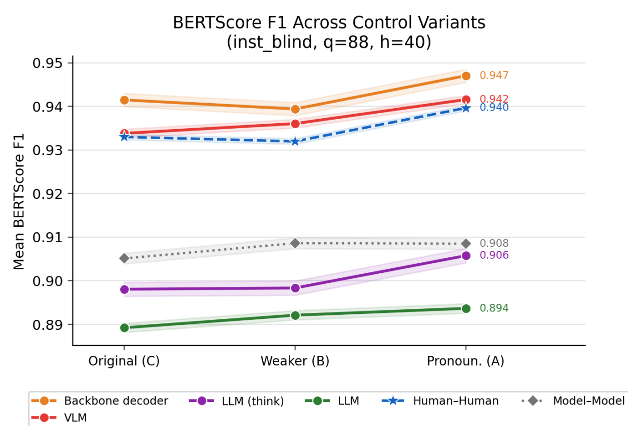

Figure 13: Human–model BERTScore F1 across control variants.

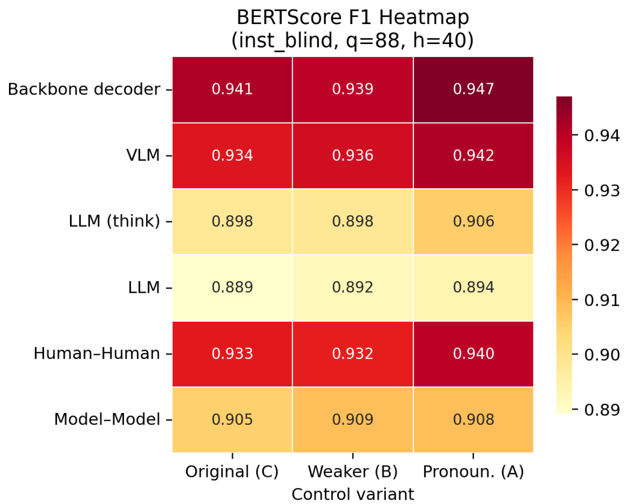

Figure 14: BERTScore F1 agreement heatmap (group × variant, HM pairs).

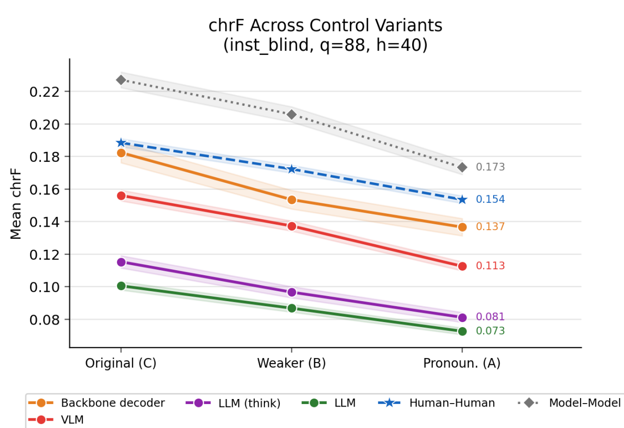

Figure 15: Human–model chrF across control variants.

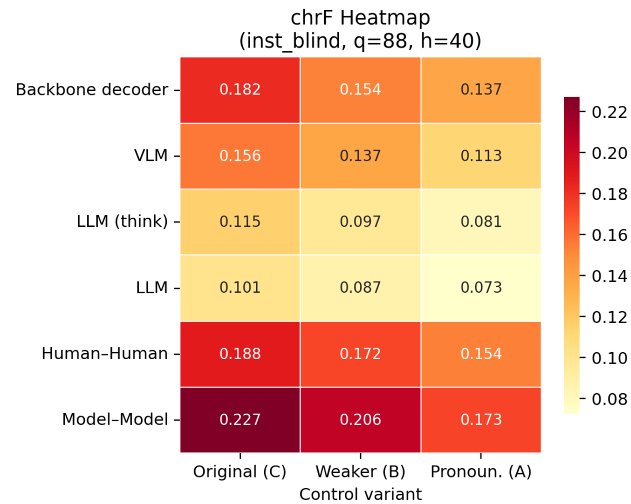

Figure 16: chrF agreement heatmap (group × variant, HM pairs).

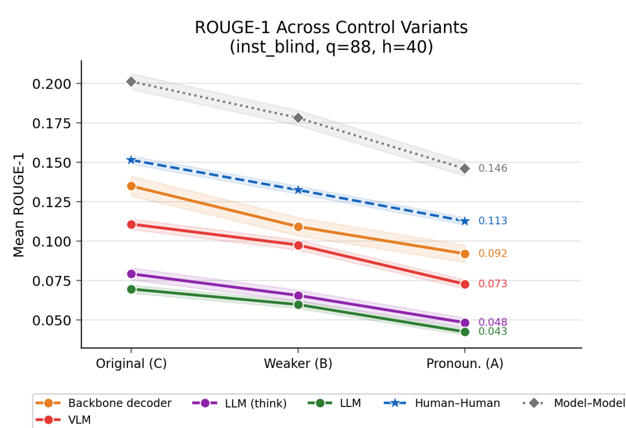

Figure 17: Human–model ROUGE-1 across control variants.

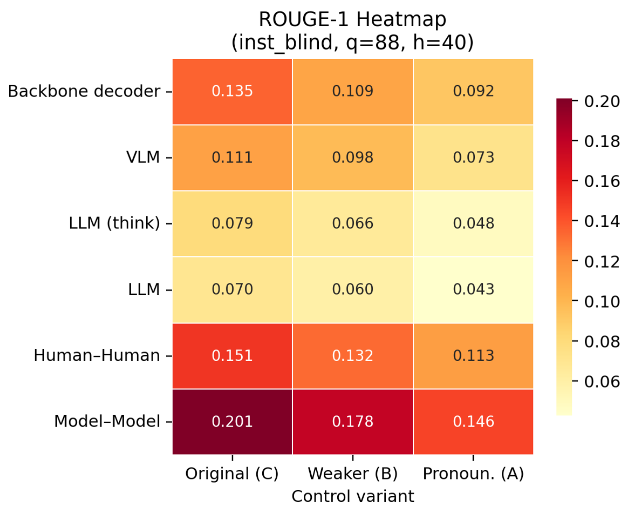

Figure 18: ROUGE-1 agreement heatmap (group × variant, HM pairs).

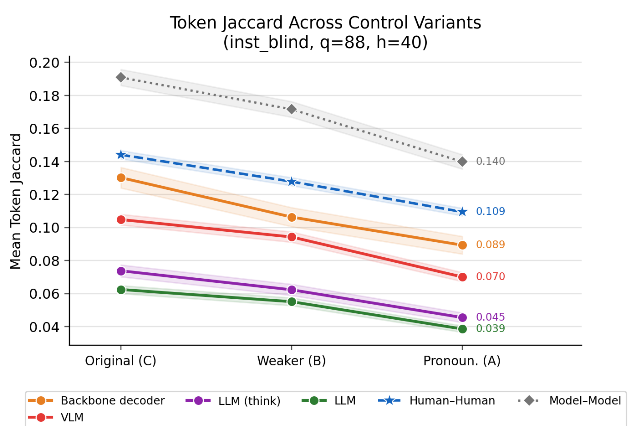

Figure 19: Human–model token Jaccard across control variants.

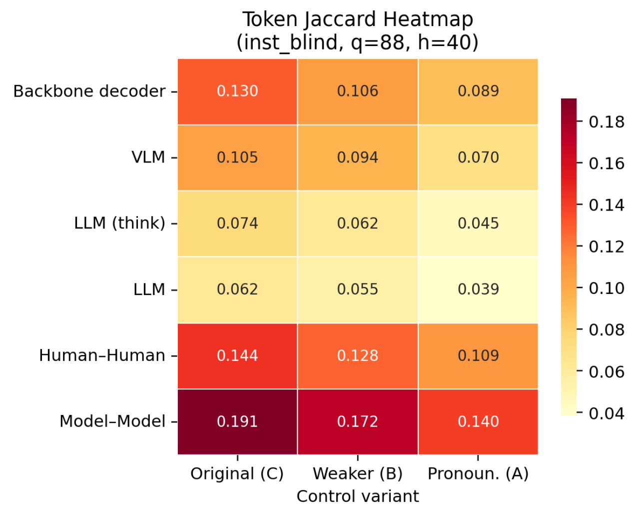

Figure 20: Token Jaccard agreement heatmap (group × variant, HM pairs).

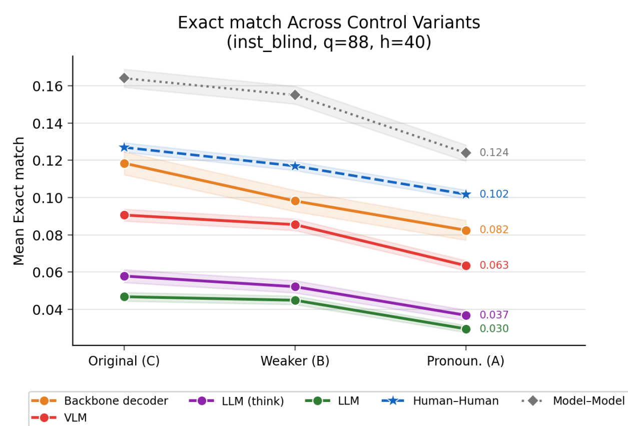

Figure 21: Human–model exact-match rate across control variants.

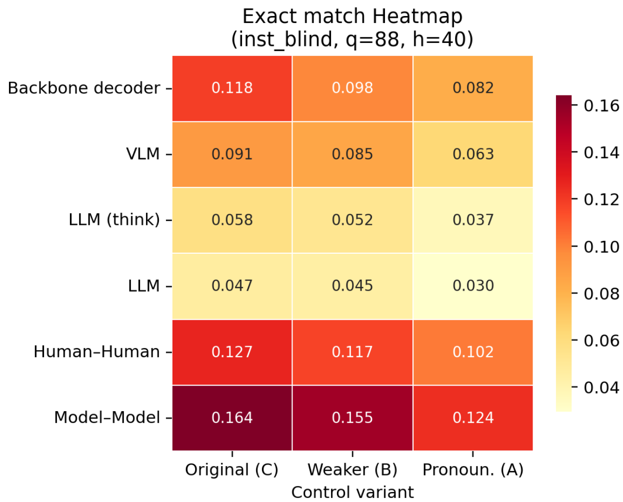

Figure 22: Exact-match heatmap (group × variant, HM pairs).

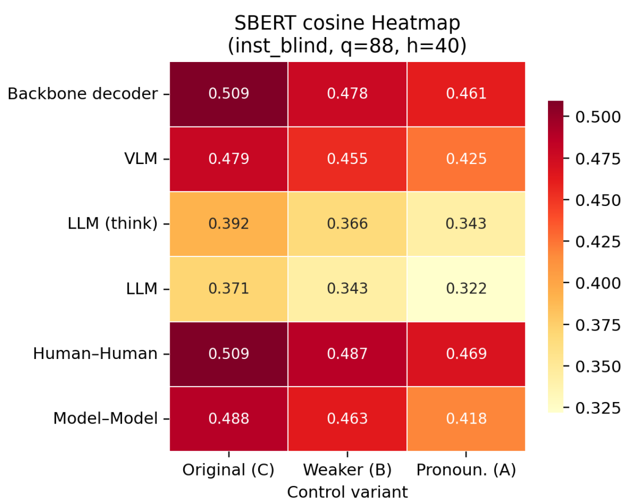

Figure 23: SBERT cosine agreement heatmap (group × variant, HM pairs). Colour scale shared with Figure 3.

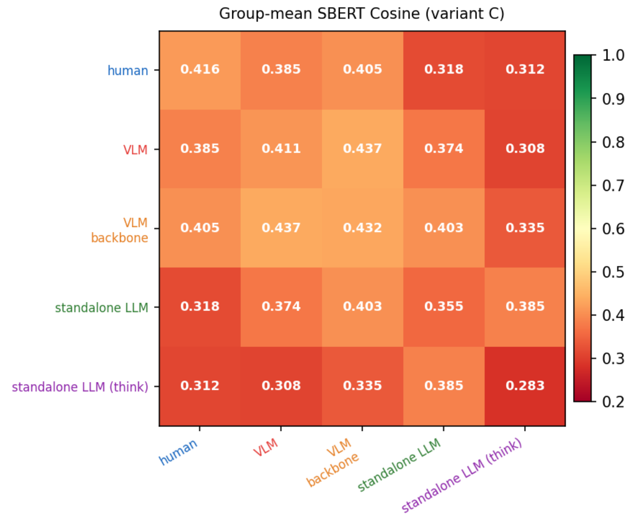

Figure 24: SBERT cosine inter-rater agreement — group-mean matrix (variant C).

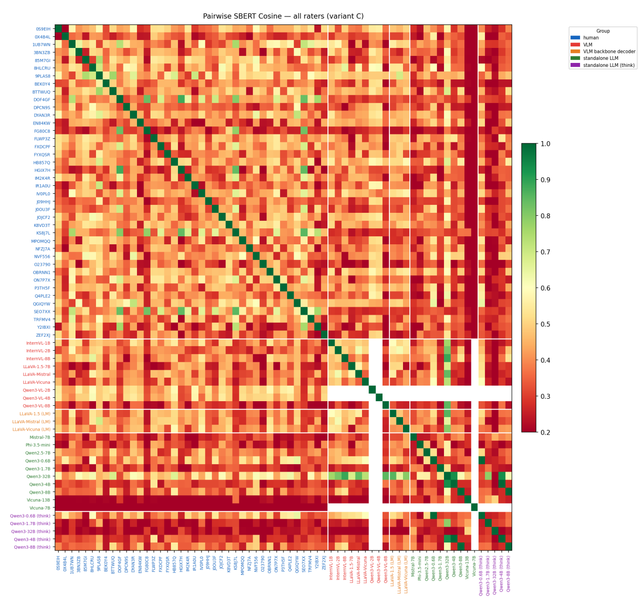

Figure 25: SBERT cosine inter-rater agreement — full per-rater matrix (variant C).

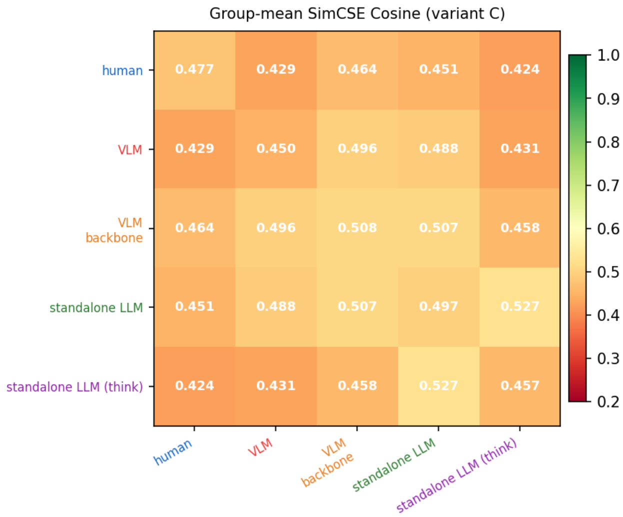

Figure 26: SimCSE cosine inter-rater agreement — group-mean matrix (variant C).

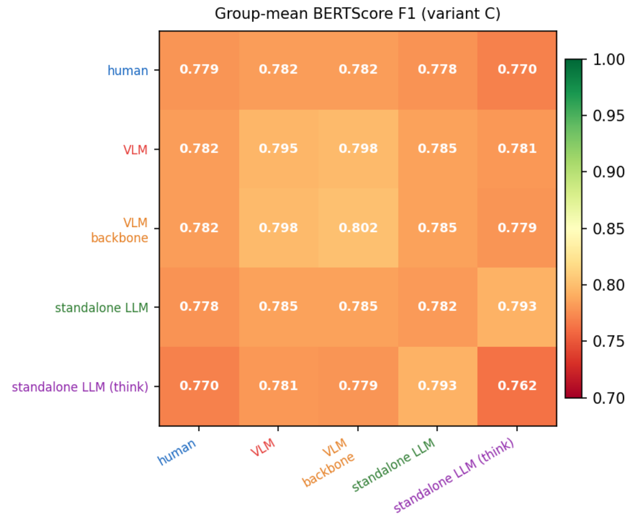

Figure 28: BERTScore F1 inter-rater agreement — group-mean matrix (variant C).

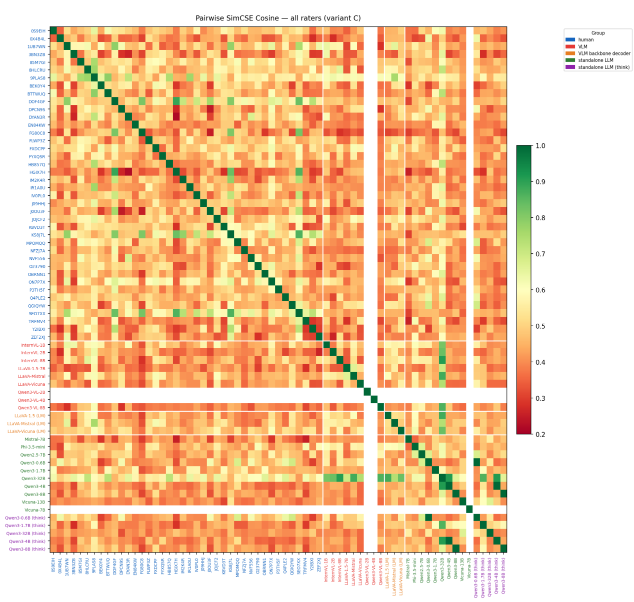

Figure 27: SimCSE cosine inter-rater agreement — full per-rater matrix (variant C).

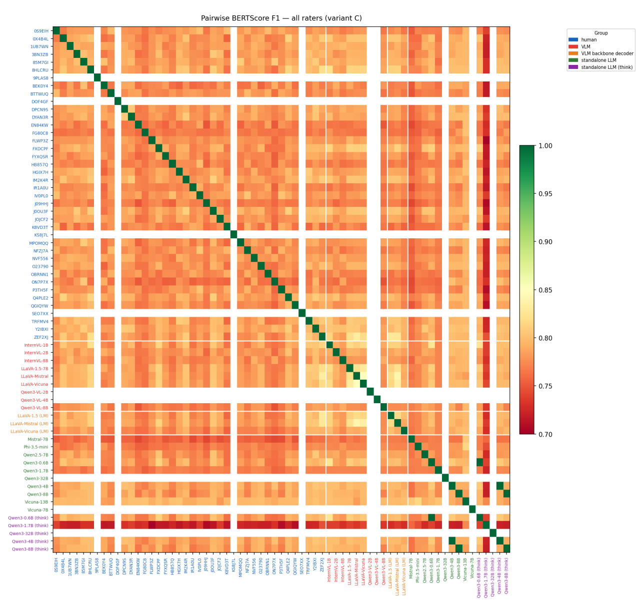

Figure 29: BERTScore F1 inter-rater agreement — full per-rater matrix (variant C).

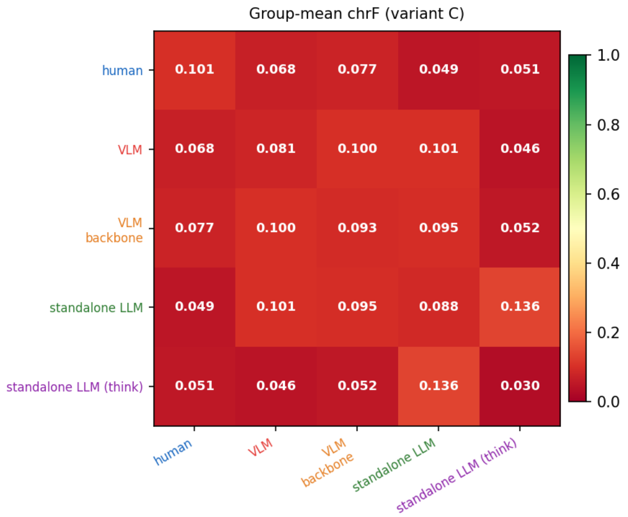

Figure 30: chrF inter-rater agreement — group-mean matrix (variant C).

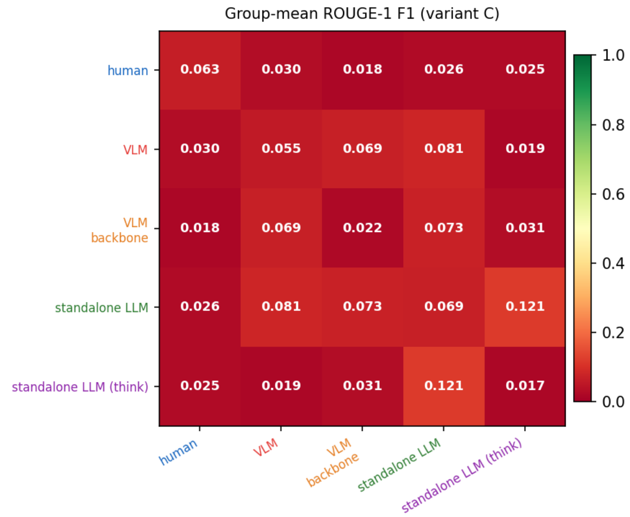

Figure 32: ROUGE-1 F1 inter-rater agreement — group-mean matrix (variant C).

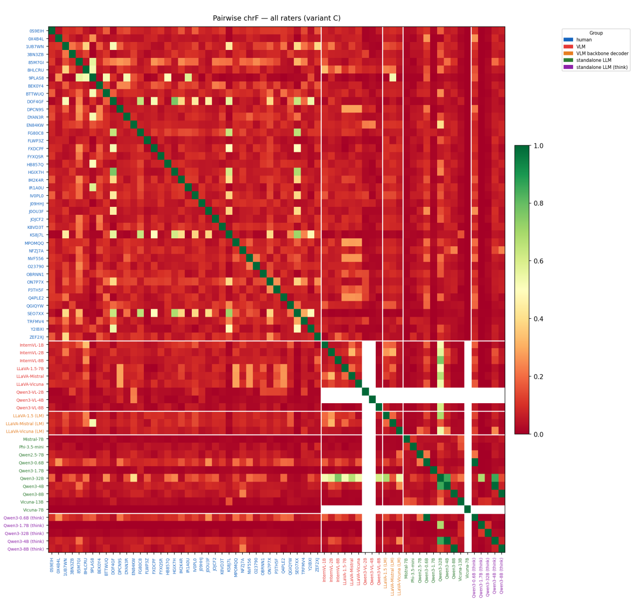

Figure 31: chrF inter-rater agreement — full per-rater matrix (variant C).

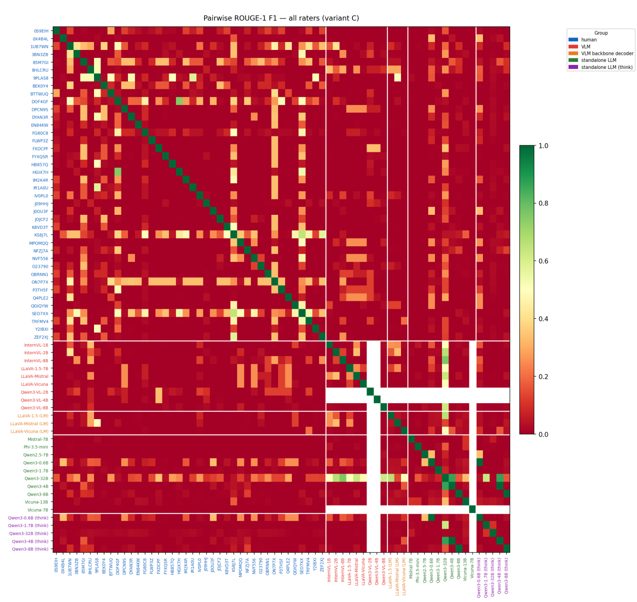

Figure 33: ROUGE-1 F1 inter-rater agreement — full per-rater matrix (variant C).

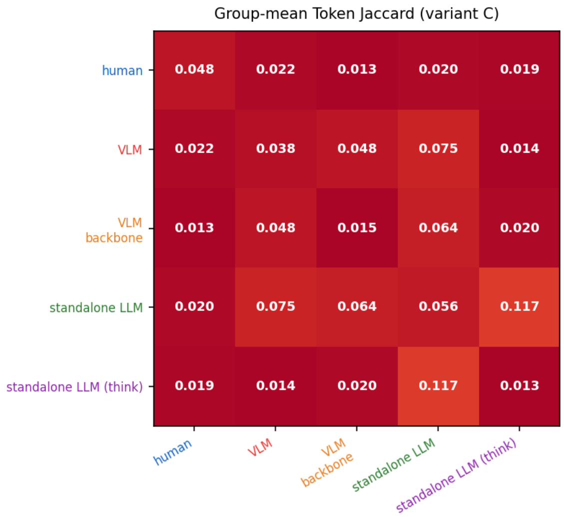

Figure 34: Token Jaccard inter-rater agreement — group-mean matrix (variant C).

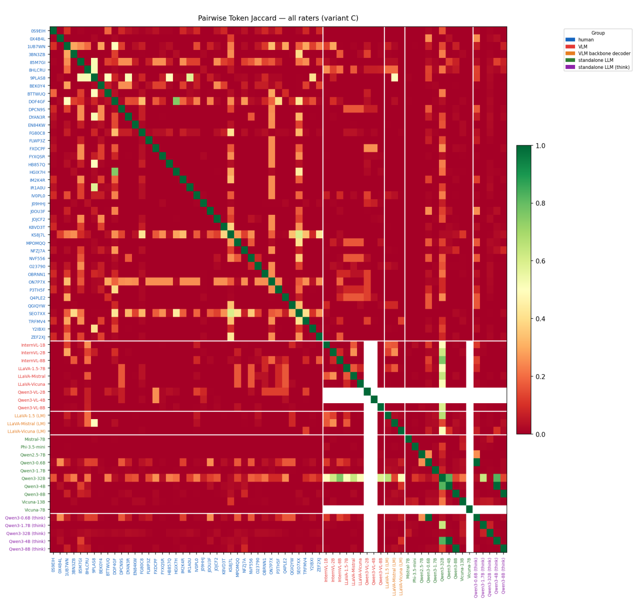

Figure 35: Token Jaccard inter-rater agreement — full per-rater matrix (variant C).

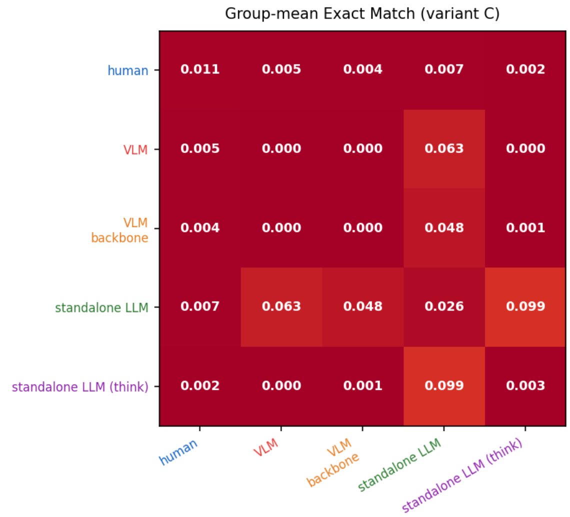

Figure 36: Exact-match inter-rater agreement — group-mean matrix (variant C).

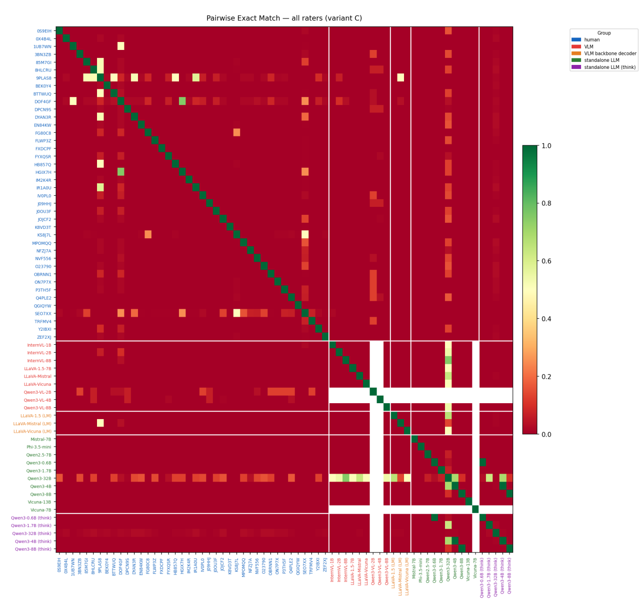

Figure 37: Exact-match inter-rater agreement — full per-rater matrix (variant C).